# Security-Aware Workload-State Counterfactual Verification and Nodal Net-Value Settlement for Spatially Coupled AI Data Centers

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***Abstract*—Artificial-intelligence (AI) data centers can respond to demand-response (DR), but a meter-only baseline cannot show that credited work was committed before an event or distinguish local reduction from spatial rebound. This paper develops an auditable submit–execute–settle verifier that carries a declared workload state into signed nodal settlement. A sparse cumulative program and an exact job-indexed start-time model enforce release, allocation runtime, energy, graphics processing unit (GPU) nameplate, contiguity, and regional-capacity constraints. A validation-fitted simplex uses total and daily conditional-value-at-risk (CVaR) budgets; a submission-masked gate and a fixed/flexible power-conversion certificate keep locked event-execution telemetry out of contract formation. On the locked 54-day mechanism-isolation panel, the total-plus-daily-CVaR profile attains normalized root-mean-square error (nRMSE) 0.324, false credit 1.787 MWh/day, and F1 score $F_1 = 0.451$; the total-only fit reaches 0.318, 1.795 MWh/day, and 0.464, making the accuracy–credit trade-off explicit rather than hiding it in an aggregate score. The complete scheduler population has 75,326 submissions and 71,128 post-event execution matches; its indexed witness contains 13,198,247 job–slot starts with an energy residual below $10^{-15}$ MWh. A binding 79-job declaration panel remains feasible under a 0.0006648-MW regional cap, and the coupled witness is replayed over all 37 finite RTS-24 contingencies. Fixed facility demand is separate and the raw $q_{99}$ endpoint is retained, so payment is conditioned on the stated workload, information, and network conditions rather than on co-located utility telemetry.**

**Index Terms—**AI data centers, counterfactual verification, demand response, security-constrained dispatch, nodal settlement.

## I. Introduction

The growth of large-model training and inference is converting data centers from nearly static industrial loads into large, software-defined grid resources. Interactive inference is latency constrained, elastic inference can be delayed briefly, and batch GPU work can be deferred for hours or migrated between regions. These capabilities enable demand response (DR), but they also invalidate the usual interpretation of a meter decrease. A reduction at one facility can be followed by delayed consumption or simultaneous rebound at another node. Paying the positive local decrease can therefore reward a redistribution that creates little or negative network value. Public traces already expose multi-week bursty workload arrivals and tens of thousands of scheduler jobs, while facility telemetry reports GPU energy at a different measurement boundary [1], [2].

Conventional DR measurement and verification estimates the unobserved no-event demand from historical meter data. Arithmetic historical baselines remain common in market practice under published market guidance [3], despite known bias and manipulation incentives [4]. Regression-based facility models quantify baseline uncertainty but do not encode a computing workload ledger [4]. More broadly, DR research establishes the economic role of responsive load, yet a meter-only estimator cannot determine whether queued jobs could actually have produced its predicted trajectory.

Data-center scheduling studies provide this physical state: temporal deferral and geographical balancing reduce energy cost and peak demand [5], [6]. Data-driven flexibility and security-constrained coordination optimize spatial–temporal dispatch [7], [8]. Balancing-market and clean-computing studies provide complementary dispatch contexts [9], [10]. Virtual-link clearing and space–time remuneration model geographical and temporal transfers [11], [12]. Receding-horizon frequency regulation retains multi-site state and future arrivals [13]. These methods optimize operator-controlled schedules; verification of an independently submitted DR counterfactual against an auditable workload ledger remains open. Recent trace and aggregator studies likewise leave ledger-to-settlement coupling open [14], [15], while cross-regional dispatchable-capacity and coupled-regulation models provide no provenance-bound certificate [16], [17]. Whole-facility measurements also require a bottom-up GPU-to-facility conversion; calibration, benchmark scaling, and network placement are therefore kept as separate evidence layers.

The second gap is electrical. Geographic workload control is often valued with energy prices or gross megawatts, although nodal marginal value changes when congestion or a contingency binds. Direct-current (DC) power flow and linear distribution factors provide transparent sensitivities [18], [19]; N–1 dispatch imposes credible post-contingency limits through line-outage distribution factors (LODFs) [20]. No data-center DR verifier jointly certifies workload feasibility, signed spatial response, and security-aware network value. Alternating-current (AC) feasibility is a separate nonlinear check. Public traces support observational replay but not a causal utility event, so measured execution, counterfactual construction, and network valuation must remain separate.

This paper makes three power-system contributions. 1) It defines a typed submit–execute–settle evidence chain: an exact job-indexed counterfactual is formed only from scheduler declarations (release, runtime, requested GPUs, and declared energy entitlement), whereas the execution ledger is retained for post-event scoring and the settlement ledger is generated only after the coupling certificate passes. The indexed witness is independently rechecked before its regional profile is valued in the N–1 model. 2) It formulates a workload-state verifier with an explicit information boundary and a validation-fitted total-plus-daily-CVaR risk fit, while keeping the pointwise false-credit cap in a separately auditable payment profile. A separately parameterized event policy and a binding-capacity stress cohort test whether the declaration-based feasible set remains nonempty when regional limits, service floors, and declared windows are jointly active. 3) It develops signed space–time nodal remuneration and a payment-activation rule over held-out power-conversion factors: the raw $q_{99}$ endpoint is retained under the declared fixed/flexible payment-network scale, while the independently reconciled flexible-only capacity factor is reported as a planning diagnostic and never used to clip that endpoint. The central result is a dimension-preserving coupling invariant that carries one declared workload state into physical verification, network valuation, and settlement; the individual linear-program (LP), dispatch, and allocation primitives are standard and are not claimed as new in isolation. Because public traces do not identify utility geography, spatial conclusions are conditional on the declared feature- stratified trace scenario and the complete permutation/penetration panel. The certificate assumes trusted scheduler and telemetry provenance: a digest detects post-commitment record changes but cannot authenticate semantically false fields.

The remainder of the paper is organized as follows. Section II defines the spatial workload and network settlement problem. Section III presents the verifier, security-aware value rule, and allocation mechanism. Section IV specifies the data, baselines, and locked protocol. Section V reports the core counterfactual, execution-replay, job-level, security, payment, continuous-horizon, and provenance experiments. Section VI concludes with the deployment boundary and future validation requirements.

## II. System Model and Problem Formulation

### A. Spatial workload state

Let $\mathcal{S}$, $\mathcal{K}$, $\mathcal{D}$, and $\mathcal{T}$ denote source regions, workload classes, destination data centers, and 15-minute intervals; the symbol $h$ is reserved for a calendar day in statistical panels. The measured ledger supplies arrival energy $a_{skt}$ for source $s$, class $k$, and release interval $t$. The decision $x_{skd\tau} \geq 0$ is the energy served at destination $d$ in interval $\tau$. Its cumulative state is

$$y_{skt} = \sum_{\tau=0}^{t} \sum_{d \in \mathcal{D}} x_{skd\tau}. \tag{1}$$

With cumulative arrivals $A_{skt} = \sum_{\tau=0}^{t} a_{sk\tau}$ and deadline $D_k$, release and deadline feasibility are

$$y_{skt} \leq A_{skt}, \tag{2}$$

$$y_{skt} \geq A_{sk,t-D_k}, \quad t \geq D_k. \tag{3}$$

The state recursion equals service, terminal service equals total arrivals, and $\sum_{s,k} x_{skdt} \leq \bar{P}_d \Delta t$. Facility demand is

$$p_{dt} = P_d^{\text{fix}} + \Delta t^{-1} \sum_{s,k} x_{skdt}. \tag{4}$$

For rolling window end $T^+$ and final event-day release $T^0$, the terminal state is not forced to consume unexpired future work. Instead,

$$y_{skT^+} = A_{sk,\max\{T^0, T^+ - D_k\}}, \tag{5}$$

which completes every job due inside the window and every event-day arrival, while retaining later real arrivals in the release constraints. The window starts $D_{\max} + 96$ intervals before the event day and ends $D_{\max}$ intervals after it; omitted earlier work must therefore expire before the evaluated day. Equations (2) and (3) prevent early execution and late completion, while (4) maps the feasible service state to the facility meter. The variable $x$ is an aggregate, preemptive service envelope: it is valid for checkpointable batch work and does not by itself assert a nonpreemptive placement for every GPU job. For the measured MIT replay, each joined job is also retained with its contiguous execution interval, runtime, GPU count, and native average power; Experiment 14 checks that witness independently. This two-level construction keeps counterfactual optimization sparse while making the task-level scope explicit. For the counterfactual task-level check, let $j \in \mathcal{J}_{\text{sub}}$ index a submitted scheduler record, $r_j$ denote its submitted release, and let $d_j$ be the service-window endpoint. Slurm "timelimit" is an allocation run-time declaration; it is converted to $R_j = \lceil \text{timelimit}_j / \Delta t \rceil$ slots and combined with a precommitted queue allowance $B = 96$ slots, so $d_j = r_j + R_j + B$. The unlimited Slurm sentinel is mapped to the precommitted horizon $H_\infty = 128$ slots before adding $B$; finite declarations are retained exactly. The 0.001-MW-per-GPU nameplate is an audit-only feasibility precheck: if a declared window cannot serve $\widehat{E}_j$, the run fails closed and the window is not enlarged. Thus $d_j$ is available at submission and never uses the observed completion time. Let $\widehat{E}_j$ be the training-calibrated declared service entitlement, $s_j$ its submit-time scenario region, and $g_j^{\text{req}}$ its requested GPU count. The indexed service variable $u_{j\tau}$ is the amount of energy assigned to job $j$ in interval $\tau$. To make the witness executable, service is a single contiguous fixed-rate block. Define $c_j = g_j^{\text{req}} \bar{p}_{\text{GPU}} \Delta t$, $K_j = \lceil \widehat{E}_j / c_j \rceil$, and the terminal-slot remainder $e_j^{\text{rem}} = \widehat{E}_j - (K_j - 1)c_j$. For each admissible start $\theta \in \Theta_j = \{r_j, \ldots, d_j - K_j\}$, a binary $z_{j\theta}$ selects one block. With $u_\tau^{j\theta} = c_j \mathbf{1}\{\theta \leq \tau < \theta + K_j\} - (c_j - e_j^{\text{rem}})\mathbf{1}\{\tau = \theta + K_j - 1\}$, the exact start-time MILP is

$$\begin{aligned}
&\sum_{\theta \in \Theta_j} z_{j\theta} = 1, \qquad z_{j\theta} \in \{0,1\}, \\
&u_{j\tau} = \sum_{\theta \in \Theta_j} u_\tau^{j\theta} z_{j\theta}, \qquad \sum_{j : s_j = d} u_{j\tau} \leq \bar{P}_d \Delta t, \\
&\min_z \sum_{j, \theta \in \Theta_j} z_{j\theta} \sum_{\tau} \left[ w_{\text{batch}}(\tau - r_j) + r^{\text{DR}} \mathbf{1}\{\tau \in \mathcal{E}\} \right] u_\tau^{j\theta}.
\end{aligned} \tag{6}$$

This exact construction enforces entitlement, contiguity, the GPU nameplate, and regional capacity at every slot. Experiment 19 enumerates every admissible start for all 75,326 submitted records; Experiment 25 solves the same model with a binding regional row. The measured $[t_j^{\text{start}}, t_j^{\text{end}}]$ interval is opened only for the independent native replay after the counterfactual is fixed: 71,128 execution matches never constrain $z$, $u$, or $d_j$. For the job-to-network certificate, the indexed witness is mapped without a second optimization. Let $\mathcal{J}_{skd}$ be the ledger partition whose source, class, and declared destination labels are $s, k, d$; Experiment 19 uses the native assignment $d = s_j$ and therefore has no outcome-dependent migration:

$$x^{\text{job}}_{skd\tau} = \sum_{j \in \mathcal{J}_{skd}:\, r_j \le \tau < d_j} u_{j\tau}, \tag{7}$$

$$p^{\text{job}}_{d\tau} = P^{\text{fix}}_d + \Delta t^{-1} X^{\text{job}}_{d\tau}, \qquad X^{\text{job}}_{d\tau} = \sum_{s,k} x^{\text{job}}_{skd\tau}. \tag{8}$$

Aggregate migration remains a matched-workload control; (8) is the indexed replay profile. Experiment 22 reconstructs $X^{\text{job}}$ from every stored $u_{j\tau}$, rechecks the indexed primal, and verifies equality before network valuation, so no separately optimized aggregate trajectory receives a network value. Workload deferral and geographic assignment follow [7], [8], while rolling-state control follows [13]; waiting and migration costs are $w_k(\tau - t)x_{skd\tau}$ and $m_{sd}x_{skd\tau}$, with service-time substitution differing only by a fixed-arrival constant.

For provenance, the submit-time ledger and post-event execution ledger have separate canonical digests. Each submitted record is represented by the scheduler-only tuple and its digest in

$$\begin{aligned} \ell^{\text{sub}}_j &= (\text{id}_j, t^{\text{submit}}_j, \text{timelimit}_j, q_j, \kappa_j, \sigma_j), \\ \mathcal{L}_{\text{sub}} &= \text{sort}\{\ell^{\text{sub}}_j : j \in \mathcal{J}_{\text{sub}}\}, \\ h_{\text{sub}} &= \text{SHA256}(\text{serialize}(\mathcal{L}_{\text{sub}}))\,. \end{aligned} \tag{9}$$

where $q_j$, $\kappa_j$, and $\sigma_j$ denote the requested-resource, class, and scheduler-state fields. The execution ledger separately stores start/end times, DCGM energy, and telemetry counts; its one-to-one join, release–execution order, and raw-to-join energy equality are checked only for post-event reconciliation. The Secure Hash Standard [21] and canonical ordering make $h_{\text{sub}}$ in (9) independent of execution telemetry. Experiment 16 reconciles the measured envelope with the 118-MW nameplate fixed before the validation/test split.

The verifier treats scheduler/DCGM services as authoritative and assumes the digest is committed before the event decision. It cannot establish that a deadline, region, class, or energy field was truthful before signing; that requires upstream attestation and lies outside this optimization certificate.

### B. Counterfactual, response, and false credit

For event set $\mathcal{E} \subset \mathcal{T}$, $p^0$ is the no-event counterfactual, $p^{1,\text{sim}}$ is the workload-optimized event trajectory, and $p^{\text{obs}}$ is the independently measured MIT/DCGM execution trace. Because no utility event label is present, $p^{\text{obs}}$ is an observational alignment target and never a causal outcome or pre-decision input. In mechanism-isolation rows $r = p^0 - p^{1,\text{sim}}$ is signed, so rebound is retained. For source $z \in \{\text{obs}, \text{sim}\}$, submitted and true credit are $\hat{r}^{(z)} = [\hat{p}^0 - p^{(z)}]_+$ and $r^{(z)} = [p^0 - p^{(z)}]_+$; source-specific false credit is

$$F^{(z)} = \Delta t \sum_{d,t \in \mathcal{E}} [\hat{r}^{(z)}_{dt} - r^{(z)}_{dt}]_+, \qquad z \in \{\text{obs}, \text{sim}\}. \tag{10}$$

The source-specific quantity in (10) is evaluated separately for the simulated and observed trajectories. The simulated source defines the validation epigraph; the observed source is recomputed only in the locked replay. For nonnegative true credit, $p^{(\text{sim})} + r^{(\text{sim})} = \max\{p^0, p^{(\text{sim})}\}$, an audited identity rather than a new meter trajectory. The gross forecast credit is not a transfer: before the event, the operator freezes $p^{\text{con}}$, and after meter closure the payable response is the pointwise intersection

$$Q^{\text{pay}} = \Delta t \sum_{d,t \in \mathcal{E}} \min\big\{[\hat{p}^0_{dt} - p^{\text{obs}}_{dt}]_+, [p^{\text{con}}_{dt} - p^{\text{obs}}_{dt}]_+\big\}\,. \tag{11}$$

Equation (11) uses only the submitted baseline, frozen contract, and closed meter. The trace-anchored $p^0$ is an offline over/underpayment diagnostic and never enters fitting, gate decisions, workload optimization, or transfer formation. The study reports gross, payable, oracle over/underpayment, nRMSE, bias, precision, recall, and $F_1$.

### C. Strategic reference scheduling

Consider an arithmetic mean of $H = 10$ reference days, event probability $q$, and DR price $\pi^{\text{DR}}$. If reference day $j$ adds event-window service $\delta R_j$, the expected payment change is

$$\Delta\Pi = \frac{q\pi^{\text{DR}}}{H} \sum_{j=1}^{H} \delta R_j - \sum_{j=1}^{H} \Delta C_j. \tag{12}$$

Let $c'_j$ be the right derivative of the minimum operating cost with respect to an event-window minimum-service constraint on day $j$. Linear-program sensitivity [22] gives the following exact local condition:

$$\frac{q\pi^{\text{DR}}}{H} > \min_j c'_j \quad \Longleftrightarrow \quad \exists \text{ a strictly profitable local deviation.} \tag{13}$$

At a degenerate optimum the dual at the current service level can be zero even when the right-direction marginal cost is positive. Experiment 1 therefore certifies $c'_j$ with two independent right finite differences of $5\times10^{-4}$ and $10^{-3}$ MWh and a $10^{-6}$ USD/MWh agreement tolerance. The historical-baseline premise is documented in [4]; (12)–(13) are this paper's workload-specific specialization.

### D. Network value

Let $M$ be the data-center-to-bus incidence matrix and $p_t = p^{\text{native}}_t + Mp^{\text{dc}}_t$ the nodal demand vector. For generator-to-bus map $C_g$, generator bounds, and power-transfer distribution factor (PTDF) $H$ with line ratings $\bar{f}$, the piecewise-linear security-constrained economic dispatch (SCED) value is

$$\begin{aligned} V(p) = \min_g \; & \sum_i C_i(g_i) \\ \text{s.t. } & \mathbf{1}^\top g = \mathbf{1}^\top p, \quad \underline{g} \le g \le \bar{g}, \\ & -\bar{f} \le H(C_g g - p) \le \bar{f}. \end{aligned} \tag{14}$$

The incidence mapping aggregates device demand at its connection bus, and (14) is the standard lossless DC representation [18], [19]. The realized signed grid value is

$$\Delta V = V(p^0) - V(p^1). \tag{15}$$

Positive $\Delta V$ is a credit and negative $\Delta V$ a debit; no positive-only truncation is applied.

## III. Proposed Methodology

### A. Model-based framework and matched-information pre-estimation

The state, power, and network equations in Section II are the first stage of the methodology: they define the feasible workload state before any statistical target or payment is formed. Fig. 1 shows the complete chain. Meter history and the submitted-job ledger enter a statistical pre-estimator and the workload feasible set in parallel. The pre-estimator never receives execution energy used as scoring truth. For each event day, it uses only the information available under one of two declared protocols: lagged meter/calendar features for online baselines, or the complete submitted ledger for equally informed post-event comparisons. Ridge regression and gradient boosting are used as standard matched-information comparators [23], [24]; extremely randomized trees and synthetic control provide two additional nonlinear and donor-pool comparators [25], [26]. Two stronger comparators are also solved: median-loss gradient boosting with the complete submitted ledger and a nonnegative synthetic control whose weights globally minimize pre-event absolute error over 28 donor days. To separate the effect of projection from that of pre-estimation, the complete-ledger quantile output is also passed through the same exact workload-feasible projection, with its penalty chosen by contiguous validation folds. The same scheduler declaration view is supplied to the post-event learner, the selected single projection, and the proposed convex verifier; execution energy and completion timestamps are withheld until the locked scoring pass.

### B. Sparse cumulative workload-state verifier

For statistical estimate $\widetilde{p}$ and each predeclared penalty $\rho_\ell$, the first-stage program solves

$$\min_{x,y,z}\ \mathcal{C}_{\mathrm{w}}(x)+\rho_\ell\sum_{d,t} z_{dt}\quad \text{s.t.}\quad z_{dt}\geq \pm(p_{dt}(x)-\widetilde{p}_{dt}),\ (16)$$

subject to (1)–(4). Absolute-value epigraphs are standard convex constructions [22]; using them to project a meter baseline onto an auditable workload ledger is specific to this work. Here $\mathcal{C}_{\mathrm{w}}$ denotes workload operating cost; $C_i$ denotes generator-segment cost in (14), and $V(p)$ denotes the network dispatch value. Thus (16) changes the objective, not the physical feasible set. Six globally solved schedules $p^{(\ell)}$ share identical arrivals and constraints. A seventh matched-information feasible-quantile projection is retained as an external transfer comparator and is included in the validation-only hull. Their convex combination $\bar{p}=\sum_{\ell=1}^{7}\alpha_\ell p^{(\ell)}$, $\alpha\geq 0$, $\mathbf{1}^\top\alpha=1$, remains feasible because the common scheduling set is convex.

The convex fit minimizes validation squared error plus the additive normalized exposure terms below, subject to total and daily-tail mechanism-isolation false-credit budgets; conditional value at risk (CVaR) measures the latter [27]. Let $\ell^\star$ be the single candidate with minimum worst contiguous-fold nRMSE and define $B_j^{\mathrm{cap}}=F_j^{(\mathrm{sim})}(\ell^\star)$, where $F_j^{(\mathrm{sim})}$ is obtained from the explicit submitted-minus-true credit definition above. Let $E_j=\max\{\sum_{n\in j} r_n^{(\mathrm{sim})},\epsilon_E\}$ be the validation credited-energy denominator and write $\widetilde{F}_j=F_j/E_j$ and $\widetilde{B}_j^{\mathrm{cap}}=B_j^{\mathrm{cap}}/E_j$. The fixed denominators precede the locked split; the observed-meter source is audited separately. For reserve $\beta\in(0,1]$,

$$\sum_j F_j^{(\mathrm{sim})}(\alpha)\leq \beta\sum_j B_j^{\mathrm{cap}},\tag{17}$$

$$\mathrm{CVaR}_{0.75}\Big(\widetilde{F}_j^{(\mathrm{sim})}(\alpha)\Big)\leq \beta\mathrm{CVaR}_{0.75}(\widetilde{B}_j^{\mathrm{cap}}).\tag{18}$$

For implementation, the positive part is not differentiated through a sort. For each event sample $n$ and validation day $j$, the program introduces $f_n\geq 0$, a CVaR threshold $\nu\geq 0$, and tail slacks $u_j\geq 0$:

$$f_n\geq p_n^\alpha-p_n^{(\mathrm{sim})}-r_n^{(\mathrm{sim})},\tag{19}$$

$$F_j^{(\mathrm{sim})}=\sum_{n\in j} f_n,\tag{20}$$

$$\widetilde{F}_j-\nu-u_j\leq 0,\tag{21}$$

$$\nu+\frac{1}{0.25J}\sum_j u_j\leq \beta\,\mathrm{CVaR}_{0.75}(\widetilde{B}_j^{\mathrm{cap}}).\tag{22}$$

Here $r_n^{(\mathrm{sim})}=[p_n^0-p_n^{(\mathrm{sim})}]_+$ is the true mechanism-isolation credit. The equivalent threshold identity is checked in the implementation, but the two credit terms remain explicit in the model. The normalized daily ratio is the reported CVaR metric, while the total constraint remains in absolute MW-slot units. This epigraph is never substituted for the frozen contract profile $p^{\mathrm{con}}$ in (11). The explicit epigraph in (22) is the implementation used in all reported fits. Together with the simplex equality, box bounds on $\alpha$, and the total budget in (17), these are linear constraints. Fits that retain an axis also include its predeclared normalized exposure term in the additive objective: total false-credit exposure and daily CVaR are normalized by their reference values and assigned equal weight. This convex preference keeps the two-axis fit identifiable when a hard row is slack; the corresponding single-axis ablations remove the omitted preference along with its constraint. The resulting objective is a convex quadratic program. HiGHS first supplies a feasible epigraph point; the full epigraph is then solved with a sparse primal–dual trust-region method. The returned stationarity residual, every linear epigraph residual, the simplex equality, and all bound residuals are recomputed from the final primal point. The optimizer status and these residuals are stored for every fit. Four contiguous folds select $\beta$ using validation nRMSE, then risk ratio; held-out noninferiority is reported as an outcome and is not used as a hidden constraint. A separately reported payment-contract projection targets $\bar{p}$ while imposing, for every event sample,

$$p_{dt}^{\mathrm{safe}}\leq p_{dt}^{(\ell^\star)}.\tag{23}$$

The pointwise cap in (23) is the upper side of the contractual credit band; it is evaluated on every event slot before settlement and is not substituted for the scientific risk profile.

*a) Two-sided credit band.:* The upper envelope alone would admit an arbitrarily conservative zero-credit profile. The method therefore imposes a lower band tied to the independently fitted convex target. Define the componentwise risk floor $p_{dt}^{\mathrm{floor}}=\min\{p_{dt}^{(\ell^\star)},\bar{p}_{dt}\}$, where $\bar{p}$ is the validation-fitted convex target before the final workload projection. The event-slot constraints are

$$p_{dt}^{\mathrm{floor}}-\varepsilon\leq p_{dt}^{\mathrm{safe}}\leq p_{dt}^{(\ell^\star)},\qquad (d,t)\in\mathcal{D}\times\mathcal{E},\tag{24}$$

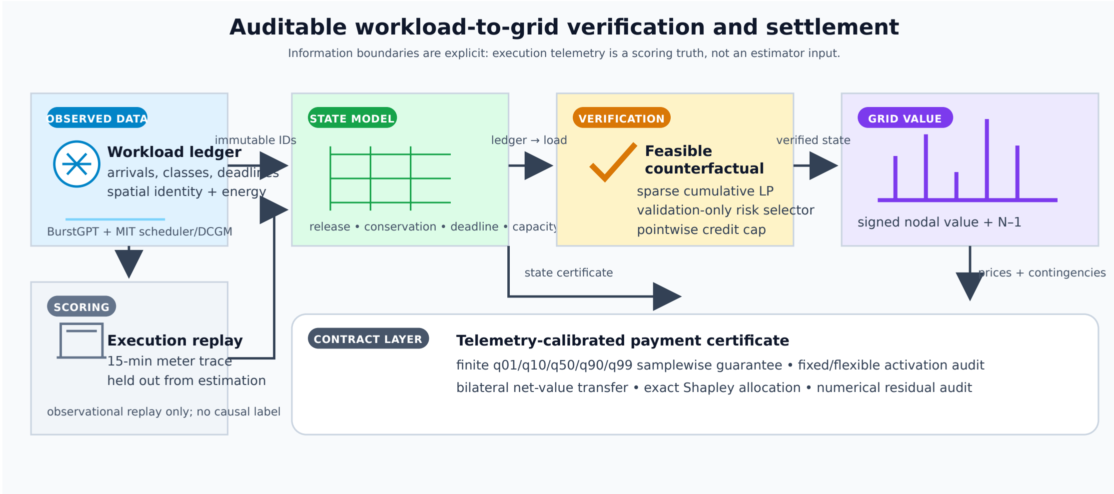


Fig. 1. Verification-to-settlement framework. Validation labels select the counterfactual risk contract; locked execution truth is used only for scoring. The right branch computes aggregate N–1 value and exact participant shares.

where $\varepsilon = 1$ MW is fixed before the locked test split. The lower bound is implemented together with the workload constraints, not as a post-processing clip. Thus the output remains an attainable schedule and can move below the single reference when the independent risk fit supports a safer trajectory. The total constraint (17) controls aggregate exposure and (18) controls the worst daily tail. The upper side is the deterministic payment cap; the lower side is a declared under-credit regularizer relative to $p^{\text{floor}}$, not a distribution-free guarantee against an arbitrary unseen meter trace. Overpayment against the realized event is handled by the contract-capped settlement rule (11). The selected single feasible projection proves nonemptiness, and the locked results report both margins and the post-event payable amount for this separate payment profile.

*b) Decision-time information boundary.:* The ex-post selector may use the complete ledger, but a gate deployment removes all post-gate arrivals before target construction and optimization. The target is refit with the matched-information metadata model on this masked ledger. A no-tariff LP freezes the contract baseline and commits 5 MWh of event service; a second LP applies the positive DR price to the same ledger and carries unexpired batch state to the next checkpoint. Reserved future jobs cannot be credited. Baseline error is computed only for the submitted contract; the tariff-bearing plan is scored as planned reduction against the closed meter and then settled by (11), keeping the two roles distinct.

To separate capacity planning from payment eligibility, the gate publishes a causal reserve envelope. For candidate quantile $\eta$ and event day $d$,

$$\widehat{a}^{\text{res}}_{sk\tau}(d;\eta) = Q_\eta(\{a_{sk\tau}(h) : h < d\}), \qquad \tau \geq t_{\text{gate}}, \tag{25}$$

Equation (25) uses temporally earlier days only. Candidates are selected on validation days under a declared false-credit budget and used only for capacity planning; no reserved job enters payment before ledger commitment.

The cumulative state makes prefix storage linear rather than quadratic in the horizon: explicit prefixes scale as

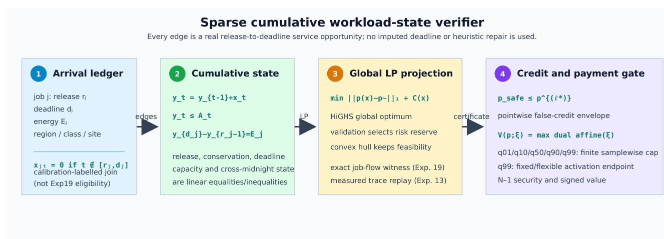


Fig. 2. Sparse cumulative verifier detail. Each admissible edge is a declared release-to-deadline service opportunity; the global projection, risk fit, and payment gate operate on the same feasible state and do not repair missing deadlines or execution records.

$O(|\mathcal{S}||\mathcal{K}||\mathcal{D}||\mathcal{T}|^2)$, whereas the cumulative matrix scales as $O(|\mathcal{S}||\mathcal{K}||\mathcal{D}||\mathcal{T}|)$. A 512-slot deadline is handled by (5) with every observed future arrival retained; no zero-arrival completion buffer is used in the final continuous-horizon certificate. The day trajectory is embedded by a lexicographic pair of linear programs: minimum L1 distance first, followed by minimum operating cost on that optimal-distance face. Hence no penalty weight trades physical fit against cost. All linear programs are solved to global optimality with HiGHS [28]. The resulting state-to-credit sequence is summarized by the framework in Fig. 1; the execution replay and interval audit are reported separately in Experiments 13–15. Fig. 2 resolves the ledger-to-LP path at the component level.

## C. Signed nodal and N–1 settlement

Let $\lambda^0 \in \partial V(p^0)$ be the nodal demand-value subgradient. Signed linear settlement is $(\lambda^0)^\top(p^0 - p^1)$, while (15) is the realized system-value benchmark and the payoff of an explicit bilateral avoided-cost contract; it is not relabeled as an independent system operator (ISO) market payment. The convex subgradient inequality [22] yields the global certificate

$$0 \leq (\lambda^0)^\top(p^0 - p^1) - \Delta V = V(p^1) - V(p^0) - (\lambda^0)^\top(p^1 - p^0). \tag{26}$$

Thus the linear rule globally upper-bounds exact value in the implemented polyhedral market model. Equality holds only when both endpoints share a supporting dual subgradient.

For market-compatible remuneration, let $\mathcal{W}$ contain the complete response cycle from the earliest deadline-feasible anticipatory shift through the latest recovery interval. The space–time rule is

$$\Pi^{\rm ST} = \Delta t \sum_{t\in\mathcal{W}} \sum_{d} \lambda^0_{dt}(p^0_{dt} - p^1_{dt}). \tag{27}$$

This is the nodal virtual-link remuneration structure of [11], [12], applied here to a verified counterfactual. Summing site payments reproduces (27) exactly.

Space–time energy remuneration is only one component of the contracted DR product. Let $\mathcal{D}^{\rm part}$ be the predeclared participating sites, $\mathcal{E}$ the event intervals, and $r^{\rm DR}$ the cleared service price. The signed delivered capacity service and total operator value are

$$Q^{\rm cap} = \Delta t \sum_{d\in\mathcal{D}^{\rm part}} \sum_{t\in\mathcal{E}} (p^{\emptyset}_{dt} - p^1_{dt}), \tag{28}$$

$$G = r^{\rm DR} Q^{\rm cap} + \Pi^{\rm ST}, \tag{29}$$

where $p^{\emptyset}$ is the minimum-cost no-event schedule under the identical continuous arrival window. Capacity credits and ex-post participation constraints follow established DR mechanism design [29], [30]; the signed $\Pi^{\rm ST}$ term separately debits spatial transfer and recovery. Equations (28)–(29) jointly define the signed service and operator value.

Let $x^1$ and $x^{\emptyset}$ denote the workload service schedules inducing $p^1$ and $p^{\emptyset}$, respectively. Define $C_{\rm opp} = \mathcal{C}_{\rm w}(x^1) - \mathcal{C}_{\rm w}(x^{\emptyset})$ as the participant opportunity cost and $S = G - C_{\rm opp}$ as the transferable transaction surplus. With disagreement utilities equal to zero, the symmetric Nash solution [31] activates the contract iff $S \geq 0$ and sets

$$z = \mathbb{I}\{S \geq 0\}, \quad P^{\rm B} = z\left(C_{\rm opp} + \frac{S}{2}\right), \quad u_{\rm dc} = u_{\rm op} = z\frac{S}{2}. \tag{30}$$

Thus (30) makes voluntary nonactivation the explicit outside option. For every activated contract, both utilities are nonnegative and the single transfer $P^{\rm B}$ is exactly budget balanced. The bilateral transfer is reported separately from both ISO nodal remuneration and the exact dispatch value benchmark. Unlike event-only accounting, $\mathcal{W}$ debits both pre-event load increases and delayed rebound; workload conservation gives $\sum_{d,t\in\mathcal{W}}(p^0_{dt} - p^1_{dt})\Delta t = 0$ up to solver tolerance.

Security-aware settlement augments (14) with every finite non-islanding single-line outage. If $L_{\ell k}$ is the LODF from outage $k$ to monitored line $\ell$, then

$$-\bar{f}_\ell \leq (H_\ell + L_{\ell k}H_k)(C_g g - p) \leq \bar{f}_\ell, \quad \ell \neq k. \tag{31}$$

Equation (31) is the standard preventive N–1 LODF formulation [20]; the experiment imposes all credible rows simultaneously, with no contingency screened out. Native public ratings are retained. Islanding outages are reported and excluded because a connected single-reference PTDF cannot represent separate islands, consistent with the scope of DC security models and the N–1 planning criterion [32].

### D. Scenario-robust N–1 payment certificate

The pointwise envelope controls credited energy but does not by itself certify network payment because spatially different demand vectors can activate different security constraints. Let $\mathcal{P} = \mathrm{conv}\{p^{(1)}, \ldots, p^{(L)}\}$ be the convex hull of complete workload-feasible schedules and let $p^{\rm cap} \in \mathcal{P}$ be the validation-selected single feasible projection. Its penalty $\rho_{\ell^\star}$ is selected by contiguous validation and is reported separately from the payment-target penalty. The feasible-quantile projection remains an external transfer comparator. Let $\Xi_5$ contain the 1st, 10th, 50th, 90th, and 99th percentiles of held-out measured-to-predicted GPU energy; these predeclared scenarios are supported by independent telemetry [2]. For facility profile $p^{\rm fac}_{dt}$, fixed demand $P^{\rm fix}_d$, and benchmark scale $s_{\rm dc}$, conversion factor $\xi$ gives

$$p^{{\rm dc},\xi}_{dt} = s_{\rm dc}\big(P^{\rm fix}_d + \xi[p^{\rm fac}_{dt} - P^{\rm fix}_d]\big), \qquad p^{\rm fac}_{dt} \geq P^{\rm fix}_d. \tag{32}$$

The scale is calibrated on the held-out $q_{99}$ flexible peak, so the fixed component is invariant to $\xi$ and both raw endpoints remain un-clipped activation scenarios; the flexible-only factor in Experiment 16 is diagnostic. In what follows, $Mp^{{\rm dc},\xi}_t$ is the nodal injection and superscripts $\alpha$ and "cap" apply (32) to $p^\alpha$ and $p^{\rm cap}$, respectively. For each settlement day, the payment-certified verifier defines $B^{\rm cap}_\xi = \Delta t \sum_{t\in\mathcal{E}} V_{N-1}(p^{\rm native}_t + Mp^{{\rm dc},\xi,{\rm cap}}_t)$ and solves

$$\begin{aligned}
&\mathrm{lexmin}_{\alpha,e,\eta,\{g^\xi_t\}} \quad \Bigg(\sum_{d,t\in\mathcal{E}} e_{dt}, -\eta\Bigg) \qquad (33)\\
&\text{s.t.} \quad p^\alpha = \sum_{\ell=1}^{L} \alpha_\ell p^{(\ell)},\\
&\qquad \alpha \geq 0, \quad \mathbf{1}^\top \alpha = 1, \quad 0 \leq \eta \leq 1,\\
&\qquad e_{dt} \geq \pm(p^\alpha_{dt} - p^{\rm safe}_{dt}),\\
&\qquad g^\xi_t \text{ satisfies (14) and (31),}\\
&\qquad\quad \text{with } p_t = p^{\rm native}_t + Mp^{{\rm dc},\xi,\alpha}_t,\\
&\qquad\qquad t \in \mathcal{E}, \quad \xi \in \Xi_5,\\
&\qquad \Delta t \sum_{t\in\mathcal{E}} \sum_i C_i(g^\xi_{it})\\
&\qquad\quad \leq (1-\eta) B^{\rm cap}_\xi, \quad \xi \in \Xi_5.
\end{aligned}$$

Generator costs use the same declared piecewise-linear epigraph as market settlement. Two global linear programs implement the lexicographic objective: the first obtains the minimum trajectory error, and the second preserves that optimum within solver tolerance while maximizing the common fractional margin $\eta$. Existence of the primal dispatches in (33) is equivalent to the corresponding upper bound on the optimal N–1 value; the unit weight on $p^{\rm cap}$ proves nonemptiness in every scenario. Consequently, for any $\xi \in \Xi_5$ and realized event load $p^{1,\xi}$, daily payment obeys

$$\begin{aligned}
&\sum_{t\in\mathcal{E}} \Delta t[V_{N-1}(p^{\rm native}_t + Mp^{{\rm dc},\xi,\alpha}_t) - V_{N-1}(p^{1,\xi}_t)]\\
&\leq \sum_{t\in\mathcal{E}} \Delta t[V_{N-1}(p^{\rm native}_t + Mp^{{\rm dc},\xi,{\rm cap}}_t) - V_{N-1}(p^{1,\xi}_t)]. \quad (34)
\end{aligned}$$

The common realized-cost term cancels, so the certificate requires no execution truth and holds samplewise. Both programs contain every finite non-islanding contingency. The formulation follows the standard SCED primal and linear-program epigraph foundations [19], [22]. Here $\mathcal{P}$ contains the six independently solved first-stage projections from (16) and the matched feasible-quantile projection. The reported risk-constrained verifier is the validation-fitted simplex point in this hull under the total and CVaR budgets. The separately stored payment-contract profile is projected toward that risk target under the pointwise activation cap; it is not fed back into the risk fit. The quantile schedule remains an independent transfer comparator.

The finite cap is complemented by a set-valued payment certificate. The contractual cap profile is denoted $p^{\mathrm{cap}} = p^{(\ell^\star)}$, where $\ell^\star$ is selected by contiguous validation nRMSE; the independent payment-target profile $p^{\mathrm{pay}}$ is selected separately by validation payment error and is used only as the target of the certificate. Let $p^{\mathrm{cert}} = p^{\alpha}$ denote the profile returned by the lexicographic solution of (33). For a realized counterfactual value, define $\Delta V_\xi(p) = V_{N-1}(p,\xi) - V_{N-1}(p^{1,\xi})$ and the continuous segment $p_\theta = (1-\theta)p^{\mathrm{cap}} + \theta p^{\mathrm{cert}}$, $0 \le \theta \le 1$. The induced interval is

$$\mathcal{I}_\xi = \left[\min_{0\le\theta\le1} \Delta V_\xi(p_\theta),\ \max\{\Delta V_\xi(p^{\mathrm{cap}}), \Delta V_\xi(p^{\mathrm{cert}})\}\right]. \tag{35}$$

The lower endpoint is obtained by one exact joint N–1 SCED LP with a shared segment parameter; the upper endpoint follows from convexity of the optimal dispatch value. The selected cap payment is an element of $\mathcal{I}_\xi$. The endpoints use only validation-frozen feasible profiles and the realized counterfactual needed for settlement; no oracle baseline selects or certifies the interval. Interval width is reported as uncertainty cost, while the seven-profile hull remains the finite-scenario payment certificate in (33).

### *E. Participant allocation*

For participants $N$, define $v(S)$ as signed dispatch value when coalition $S$ moves from baseline to actual demand and nonmembers remain at baseline. The exact allocation is

$$\phi_i(v) = \sum_{S\subseteq N\setminus\{i\}} \frac{|S|!(|N|-|S|-1)!}{|N|!}[v(S\cup\{i\}) - v(S)]. \tag{36}$$

This is the Shapley value [33]; the characteristic function based on signed grid value is the proposed specialization. Grouping (36) by coalition size cancels every intermediate value and gives $\sum_i \phi_i = v(N)$ when $v(\varnothing) = 0$. All 16 coalitions for four sites and all 256 coalitions for eight contractual portfolios are enumerated; no permutation estimator is used. For larger auditable contracts, exchangeable slices at the same electrical site are grouped by their member count. The exact sum retains the binomial multiplicity of every labeled coalition and evaluates only $\prod_g (m_g + 1)$ distinct count states. This symmetry reduction remains the Shapley value and is evaluated through 20 participants.

### *F. Analytical guarantees*

The following certificates make the logical scope explicit. First, under the interior and differentiability assumptions of (12), a feasible reference-day perturbation is strictly profitable if and only if $q\pi^{\mathrm{DR}}/H > \min_j c'_j$ in (13); Experiment 1 obtains the right-hand-side duals and checks the independently solved 56-cell strategic grid. The workload set is linear and therefore convex. Every simplex schedule and the two-sided envelope in (24) are feasible, and for any execution $c$,

$$[p^{\mathrm{safe}}_{dt} - c_{dt}]_+ \le [p^{(\ell^\star)}_{dt} - c_{dt}]_+,$$
$$[c_{dt} - p^{\mathrm{safe}}_{dt}]_+ \le [c_{dt} - p^{\mathrm{floor}}_{dt}]_+ + \varepsilon.$$

Summation gives the false-credit and under-credit bounds without using locked outcomes.

Second, the base-case and stacked N–1 dispatch values are convex and piecewise affine in nodal demand. The subgradient inequality is exactly (26), with $0 \le \Delta V_{\mathrm{lin}} - \Delta V \le \|\lambda^1 - \lambda^0\|_* \|p^1 - p^0\|$; positive-only settlement has no such identity. In (33), each embedded dispatch is a feasible witness, so subtracting the common realized cost gives (34) for every $\xi \in \Xi_5$. The dual representation also makes $V_{N-1}(p;\xi)$ convex in a continuous conversion factor, so the worst case on $[\xi_{\mathrm{L}}, \xi_{\mathrm{U}}]$ is attained at an endpoint. The lower endpoint of (35) is the exact joint-LP minimum over the declared affine workload segment and convexity makes its endpoint maximum exact; Experiments 9 and 15 use independent 40- and four-segment models.

Finally, collision resistance of SHA-256 makes post-commit changes to the scheduler-only tuple detectable in $h_{\mathrm{sub}}$, while execution joins check ordering and energy conservation only after the event. An optimum of (6) gives each submitted job its entitlement within its submit-time window, enforces site capacity and the requested-GPU nameplate, and does not rely on aggregate rounding; contiguous execution is audited separately. With $B_{dj} = \mathbf{1}\{s_j = d\}$ and bus map $M$, the job, aggregation, and mapping residuals are

$$r^{\mathrm{job}}_j = \sum_{\tau=r_j}^{d_j-1} u_{j\tau} - \widehat{E}_j,$$
$$r^{\mathrm{agg}}_{d\tau} = X_{d\tau} - \sum_j B_{dj} u_{j\tau},$$
$$r^{\mathrm{map}}_{b\tau} = p_{b\tau} - P^{\mathrm{fix}}_{b\tau} - \Delta t^{-1} \sum_d M_{bd} X_{d\tau}.$$

If their typed norms are at most $\varepsilon$, the dispatched profile is the same indexed witness up to that tolerance; at zero residual an independently optimized aggregate cannot receive settlement value. Experiment 22 checks these residuals before N–1 valuation and preserves $\sum_i \phi_i = v(N)$.

## IV. Experimental Design

### *A. Data, preprocessing, and information boundary*

The complete data inventory is as follows. BurstGPT supplies 5,188,507 successful requests and token service [1]; MIT Supercloud supplies scheduler submissions and independent Data Center GPU Manager (DCGM) energy. The full scheduler population contains 75,326 valid submissions and 71,128 positive-energy execution matches, of which 68,664 start in the common 121-day tensor used by Experiments 1–13 [2]. Submission time defines arrivals and measured intervals define scoring truth. A held-out power calibration is chronological: 44,391 jobs in the first 40 complete days are used for fitting and 50,791 later jobs are held out, yielding held-out $R^2 = 0.8331$, MAE 13.02 W, and RMSE 19.14 W. Workload scaling is also fitted only on the first 40 complete training days. The held-out per-job energy ratios define $\Xi_5$ in (33) without synthetic replacement. Experiment 14 extends the replay to the last completion and retains all 71,128 matches, whereas Experiment 19 uses every valid scheduler declaration and a training-only entitlement; the execution join is opened only for post-event scoring. Its runtime window plus the precommitted 96-slot queue allowance is a contract-window definition, not a submit-to-completion interpretation of the Slurm allocation runtime. Calibration membership is fixed by chronology and never by an outcome-filtered job identifier.

Because the traces lack utility bus locations, four-region coupling is a trace-driven benchmark: a predeclared feature-stratified round robin assigns class, GPU count, runtime, and submission time. Experiment 11 exhausts all 24 permutations, three penetrations, and two concentration controls. Experiment 18 uses fixed generator-bus electrical-role strata at native-load multiplier 0.90, so its cross-network result is conditional on the declared trace-to-bus scenario. Held-out energy ratios span 0.5367–4.7557. Equation (32) carries the fixed 6-MW/site component separately and calibrates the payment network scale on $q_{99}$; both raw endpoints remain activation-eligible in that declared payment model. The separate flexible-only 0.8901 capacity-safe factor is retained as a planning diagnostic, not an endpoint clip. PGLib and MATPOWER/PYPOWER provide topology, ratings, and costs [19], [34]; RTS-24 follows the published case [35].

### B. Baselines, parameters, and inference

Experiment 2 compares eight meter/statistical baselines with the complete-ledger quantile, feasible projection, synthetic control, tail-risk counterfactual, the selected single projection, and risk-constrained verifier. payment-certified verifier is evaluated separately in Experiments 9 and 15 because its hull and N–1 objective differ. Feature equality is audited while execution truth is withheld. Deferral windows are 0, 4, and 512 slots; each declared region has a 118-MW flexible nameplate and a 6-MW fixed facility component. Six projection penalties and four risk reserves are validation-only, selected by nested contiguous folds before the locked split; normalized total-exposure and daily-CVaR preferences are fixed before the split, and four committed-response weights are fitted on earlier validation days. The gate commits 5 MWh, and payment is scored by an independent 40-segment evaluator. Moving three-day bootstrap replicates and all $2^{18}$ block-sign assignments use Holm correction [36], [37]. Sparse-system size grows from 5,760 variables/13,424 nonzeros at 96 slots to 36,480/85,488 at 608 slots (0.039–0.212 s).

The model-out check uses polar alternating-current optimal power flow (AC OPF) with active/reactive balance, voltage/reactive-generation bounds, and apparent-power ratings [19]. After the intact solve, every credible outage imposes, for each non-reference generator $i$,

$$P_{Gi}^{(k)} = P_{Gi}^{(0)}, \qquad i \notin \mathcal{G}_{\mathrm{ref}},\ k \in \mathcal{C}. \tag{37}$$

The reference generator balances active losses, while $Q_G^{(k)}$ and voltages remain recourse. Apparent-power and voltage limits are checked after each fixed-plan solve, so (37) tests one shared active plan rather than reoptimizing active schedules.

TABLE I
LOCKED 54-DAY MECHANISM-ISOLATION RESPONSE RESULTS (DAILY MEANS).

| Method | nRMSE | False MWh | Under-credit | $F_1$ |
|---|---|---|---|---|
| High-5-of-10 | 4.196 | 172.455 | 6.474 | 0.143 |
| Ridge | 2.761 | 109.817 | 5.306 | 0.233 |
| Gradient Boosting | 2.343 | 96.077 | 4.817 | 0.245 |
| Extra Trees | 3.350 | 133.961 | 6.804 | 0.171 |
| Metadata Gradient Boosting | 2.201 | 92.874 | 4.193 | 0.259 |
| Ex-post Metadata Gradient Boosting | 2.126 | 90.807 | 4.184 | 0.265 |
| Ex-post Quantile Gradient Boosting | 0.932 | 23.144 | 6.085 | 0.533 |
| Synthetic Control | 2.598 | 104.998 | 7.877 | 0.202 |
| Feasible Quantile Projection | 0.301 | 2.811 | 10.772 | 0.519 |
| Single Feasible Projection | 0.339 | 1.823 | 13.537 | 0.418 |
| Tail-Risk Feasible Counterfactual | 0.323 | 1.791 | 12.928 | 0.452 |
| **Risk-Constrained Convex Verifier** | **0.324** | **1.787** | **12.948** | **0.451** |

Non-reference bounds are fixed at the intact output $\pm 10^{-8}$ MW with a $10^{-6}$-MW residual margin; the largest recorded deviation is $2.52 \times 10^{-8}$ MW.

Experiments 1–2 test incentives and workload verification; 3–9 test signed settlement, transfer, risk, and N–1 value; 10–12 test AC/spatial transfer and continuous-cycle remuneration; 13–16 provide measured replay, the indexed witness, conversion intervals, and ledger reconciliation. Experiments 17–25 then freeze the gate information set, verify shared-plan AC feasibility, reconstruct the submitted-job witness, audit scale/coupling, replay an independent event, and enumerate all finite RTS-24 outages. Confirmatory comparisons use the same 54 locked days unless a deterministic network or ledger rule is stated.

## V. RESULTS

### A. Incentive boundary and independent verification

Experiment 1 solves the entire $8 \times 7$ DR-price/call-probability grid. The dual threshold (13) agrees with the independently solved strategic program in all 56 cells. At $150/MWh and $q = 0.35$, strategic reference scheduling inflates the event baseline by 8.223 MWh. Of 55.452 MWh credited, 44.458 MWh is unsupported by physical reduction, giving a false-credit ratio of 0.802. The profitable region follows the derived marginal boundary and does not use a fitted classifier; the complete grid and its phase diagram are retained in the experiment output.

Experiment 2 evaluates twelve methods on the locked 54-day panel (Table I). Joint total-plus-daily-CVaR gives nRMSE/$F_1$/false credit 0.324/0.451/1.787 MWh/day; total-only and CVaR-only give 0.318/0.464/1.795 and 0.323/0.452/1.791. The CVaR row binds (ratio slack $1.8 \times 10^{-8}$), the total row retains 5.55 MW-slot slack, and the joint simplex differs from CVaR-only by $L_\infty = 2.95 \times 10^{-2}$. Single projection gives 0.339/0.418/1.823, whereas the unconstrained fit gives 0.297/0.521/3.241. Validation joint/CVaR-only/total-only values are 0.427/0.430/3.141, 0.426/0.432/3.153, and 0.418/0.449/3.177; the payment contract cap is separate from this accuracy–credit trade-off.

The independent trace-meter replay is a complementary deployment check: event-window MAE is 2.673/2.873/2.878 MW for feasible quantile/tail-risk counterfactual/risk-constrained verifieron the same 54 locked days (three-day block intervals in Experiment 20). It measures closed-DCGM alignment and remains separate from the mechanism-isolation response panel.

Closest domain baselines use identical arrivals, deadlines, capacities, event slots, and locked days. The event-reward

TABLE II
BASE-CASE AND COMPLETE N–1 SETTLEMENT ERRORS (USD/DAY).

| Panel | Settlement | Trace MAE | Workload MAE | Trace median/overpay | Workload median/overpay |
|---|---|---|---|---|---|
| Base case | Nodal gross | 360.580 | 421.677 | – | – |
| Base case | Nodal exact net value | **151.719** | 455.865 | – | – |
| N–1 | Base exact | 139.221 | 167.367 | 79.79/130.42 | 121.01/107.69 |
| N–1 | N–1 linear | 2.208 | **150.461** | 0.60/1.91 | 113.43/20.38 |
| N–1 | **N–1 exact** | **0.990** | 150.955 | 0.87/0.01 | 113.43/20.18 |

and batch-flexibility equation translations give nRMSE/false-credit/$F_1$ 3.068/114.776/0.137 and 0.339/1.823/0.418, respectively. The rolling-horizon, cross-regional dispatchable-capacity, and coupled multi-service translations are solved on the same ledger; their equations, information sets, and implementation status are listed in the machine-readable baseline audit. The rolling-horizon translation follows [7], while the cross-regional and coupled- regulation translations follow [16], [17]. These are published-equation translations, not claims of reproducing unpublished software or private data; temporal-only and joint ledger controls migrate 0 and 0.206 MWh. The comparison therefore tests structural mechanisms without attributing internal control rows to the cited authors. The ablation panel contains 810 globally solved schedules (binding shares 0.310/0.948; slope 1.003).

### B. Nodal value and spatial response

Experiment 3 crosses thirteen counterfactuals with five settlement mechanisms. Table II isolates the mechanism with the trace-anchored reference and reports the locked workload rows. Exact signed nodal value has MAE \$151.719 versus \$360.580 for nodal gross response; the corresponding workload errors are \$455.865 and \$421.677. Signed netting reduces the trace error by \$208.190/day (Holm-adjusted $p = 3.89 \times 10^{-4}$), whereas uniform-to-nodal pricing changes it by $-\$1.463$ ($p = 0.0118$) and linear-to- exact value by \$0.671 ($p = 0.8026$). Workload effects are $-\$34.165$, $-\$1.191$, and $-\$0.023/day$, quantifying the mechanism-specific trade-offs on the same locked workload panel. All 5,616 interval/counterfactual instances satisfy (26); the largest violation is $2.03 \times 10^{-11}$ dollars.

Experiment 4 applies the predeclared largest gross-minus-net offset (day 87), exposing the rebound component that positive-only local accounting omits.

### C. Cross-network and N–1 security evidence

Experiment 5 evaluates 6,480 four-network outcomes; exact value improves the trace-anchored error in all 12 cells (mean \$1.913/day; eight Holm-adjusted tests below 0.05), with a \$0.050/day workload change. Experiment 6 solves 810 capacity/deadline schedules (binding shares 0.031–0.911; false credit 2.374–2.740 MWh).

Experiment 7 enumerates 16 four-site and 256 eight-site coalitions (maximum budget residual $5.68 \times 10^{-14}$ dollars) and 1,296 states in the exact 20-participant panel. Experiment 8 enforces (31) for all 37 RTS-24 non-islanding outages: trace/workload MAE is \$139.221/\$2.208/\$0.990 and \$167.367/\$150.461/\$150.955 for base exact/N–1 linear/N–1 exact, with 1.000 p.u. maximum loading.

Table IV summarizes the complete replay scope.

Experiment 9 tests (34) on 54 global programs with a validation-frozen seven-profile hull; contiguous-fold nRMSE

TABLE III
INFORMATION-BOUNDARY RESPONSE CERTIFICATES (54-DAY MEANS). THE TWO INDEPENDENT POLICIES USE A SUBMISSION-MASKED LEDGER; THE COMPLETE-LEDGER ROW IS MECHANISM ISOLATION.

| Method | nRMSE | Gross false MWh | Payable MWh | Meter underpay | Recall | F1 |
|---|---|---|---|---|---|---|
| Independent gate-committed response | 2.606 | 145.506 | 227.116 | 94.055 | 0.436 | 0.358 |
| **Independent risk-constrained response** | **1.712** | **30.036** | **38.396** | **167.306** | **0.038** | **0.054** |
| Complete-ledger risk-constrained verifier (mechanism isolation) | 0.339 | 1.823 | 10.009 | 13.537 | 0.335 | 0.418 |

selects the cap and validation N–1 MAE selects the payment target. Held-out factors are $q_{01} = 0.537$, $q_{10} = 0.669$, $q_{50} = 0.924$, $q_{90} = 1.410$, and $q_{99} = 4.756$; every certificate violation is below $9 \times 10^{-9}$ USD. For PaymentSafe/single/feasible-quantile/RiskSafe, q50 MAE/overpayment is \$41.793/34.978/36.587/35.173 and \$1.747/4.587/4.130/4.517; q99: \$217.497/182.676/191.267/183.912; \$9.450/24.695/22.242/24.369. The paired Pareto file reports moving-block 95% intervals against every comparator, keeping the cap certificate distinct from a universal MAE claim. Experiment 15 retains $q_{01} = 0.5367$ and $q_{99} = 4.7557$; the fixed/flexible conversion (32) activates both raw endpoints in the q99-calibrated model without clipping, while 0.8901 is a planning diagnostic. Unseen 0.80/1.20 MAE is \$9.303/\$13.958 (payment) and \$9.841/\$14.764 (feasible quantile), with neither factor certified; stress panels share the declared resource semantics.

Experiment 10 replaces the linear network with AC OPF [19]: all 864 connected-base and 5,400 corrective IEEE-9/14 cells are feasible, and the IEEE-9 shared-plan panel has zero active-plan deviation over 3,888 cells (0.880/0.907/0.935 p.u. at 3/6/9%). Experiment 18 extends this to four networks and 9,648 admissible outages over six snapshots, with zero voltage violation and maximum loading 1.000 p.u. Experiment 11 averages 26 assignments over three penetrations, four counterfactuals, and 54 days; 3/6/9% MAE is \$101.624/\$204.300/\$304.603 for RiskSafe, \$105.250/\$211.555/\$315.490 for single, and \$81.613/\$164.249/\$244.481 for feasible quantile per day. Experiment 16 audits 44,391/50,791 chronological training/test GPU-power observations (MAE 13.023 W, RMSE 19.137 W, $R^2 = 0.8331$); Experiment 12 reports event-window MAE \$26.937/\$15.969/\$5.325/\$5.325 and complete-cycle residual \$0.621/\$0.607/\$0/\$0 for feasible-quantile, payment-certified, RiskSafe, and single projection. The zero residuals are LP-precision accounting certificates; event-window error is predictive, all 54 contracts activate, and security sets are complete declared sets.

### D. Independent trace and deployment-boundary audits

Immutable MIT scheduler/DCGM replay gives slot MAE 3.740 versus 5.678 MW at 100% coverage and a 71,128-job residual below $10^{-17}$ MWh; no utility event label is available. Trace-meter MAE is 2.988 MW (2.673 for feasible quantile). Experiment 23 uses a submission-masked ledger, predeclared tariff, and an independent LP with no target, envelope, hull, or verifier output. Over 54 locked days, gate/independent responses score nRMSE 2.606/1.712, false credit 145.506/30.036 MWh/day, and $F_1 = 0.358/0.054$ (Table III); their nonzero trajectory differences certify information separation. Validation selects \$10/MWh, weight 1.0, and a 20-MWh/day budget; $\eta = 0.60$ is planning-only.

TABLE IV
COMPLETE DECLARED SECURITY-REPLAY COVERAGE.

| Panel | Cells and certificate |
|---|---|
| Preventive DC N–1 (RTS-24) | 54×8×37; 1.0000 p.u. |
| Shared-plan AC (IEEE-9) | 3,888; 0.880/0.907/0.935 p.u. |
| Corrective/all-outage AC | 864+5,400; 31,968 solves; all feasible |

TABLE V
EXACT INDEXED JOB-LEVEL AND BINDING-CAPACITY CERTIFICATES.

| Quantity | Value |
|---|---|
| Submitted / execution-matched jobs; admissible starts | 75,326 / 71,128; 13,198,247 |
| Native / counterfactual event energy; gross/net; rebound (MWh) | 0.194 / 0; **0.194/0.194**; 0 |
| Maximum job residual; minimum site slack (MWh) | 0; 29.486 |
| Binding stress cohort / capacity / service floor (jobs/MW/MWh) | 79 / 0.0006648 / 0.004 |
| Capacity-proportional / fixed-nameplate scale (×) | 2176.183 / 1.504 |
| Job-to-network residual; coupled event slots (MWh / count) | 0; 1,056 |
| Secure N–1 value; RTS-24 outages / cells / solves | **0.004228**; 37 / 864 / 31,968 |
| Maximum counterfactual post-contingency loading (p.u.) | 1.0000 |

### E. Indexed workload counterfactual

Experiment 19 assigns one exact contiguous start block to each of the 75,326 valid submissions. The declaration window, 0.001-MW/GPU bound, regional entitlement, and 118-MW capacity yield 13,198,247 admissible starts with typed residuals; native/counterfactual event energy is 0.194/0 MWh, with zero job-energy residual and 29.486 MWh minimum site slack (Table V). Experiment 22 replays this witness over 1,056 event slots and all finite RTS-24 outages, obtaining \$0.004228/\$0.006359/\$9.199820 raw/fixed-nameplate/capacity-proportional secure value per interval. Experiment 21 reports 2176.183× capacity-proportional stress and 1.504× fixed-nameplate scale.

Experiment 25 independently validates the declaration boundary: 71,128 of 75,326 submissions match execution, no declared window is infeasible, and its 79-job cohort binds 0.0006648 MW per region while meeting the 0.004-MWh floor with 78.125% of event-capacity slots active. Experiment 24 freezes two profiles and replays all 37 finite RTS-24 outages (864 cells; 31,968 solves), all feasible at 1.000000 p.u. (Table V).

## VI. CONCLUSION

This paper couples a provenance-bound workload verifier to signed nodal settlement. The validation-fitted total-plus-daily-CVaR profile reaches nRMSE/false credit/$F_1$ = 0.324/1.787/0.451 MWh/day, while the slot-60 replay uses only committed declarations (Table III). The indexed witness contains 13,198,247 admissible contiguous starts over 75,326 submissions, gives 0.194 MWh net reduction with zero rebound, and passes the typed RTS-24 N–1 coupling certificate. A 79-job cohort binds 0.0006648 MW while meeting the 0.004-MWh floor; fixed demand stays outside conversion, so raw $q_{99}$ remains auditable and the capacity-proportional scale is a stress transform. Deployment requires signed co-located telemetry and topology access; causal utility validation remains external.

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